

Isaac Legene

314-605-7147 | ilegene3@gatech.edu | isaac-legene.github.io

EDUCATION

Georgia Institute of Technology

M.S. in Robotics - Electrical and Computer Engineering (GPA: 4.0)

Atlanta, GA

2025 – 2027

Missouri University of Science and Technology

B.S. in Engineering Management - Industrial Engineering (GPA: 3.51)

Rolla, MO

2022 – 2024

ACADEMIC EXPERIENCE

Georgia Institute of Technology: LIDAR Lab

Graduate Researcher - Humanoid Whole-Body Loco-Manipulation and Tactile Sensing

Atlanta, GA

2025 – Present

Diffusion Policy Fine-tuning for Humanoid Loco-Manipulation via Reinforcement Learning

IEEE RA-L 2026

- Trained and fine-tuned reinforcement learning and diffusion policies for humanoid loco-manipulation in Isaac Lab.
- Designed simulation environments and robot logic for data collection and policy training.

Tactile-Based Whole-Body Manipulation via Force-Supervised Contact-Aware Planning

CoRL 2026

- Participated in the design and implementation of tools for our in-house data collection interface.
- Coordinated, collected, and parsed human demonstration and teleoperated tactile data for policy training.

Georgia Institute of Technology: College of Computing

Graduate Teaching Assistant – Machine Learning (CS 7641)

Atlanta, GA

Aug 2026 – Present

ROBOTICS AND AUTOMATION EXPERIENCE

Tesla

Automation Controls Engineering Intern

Material Flow Automation - Autonomous Mobile Robotics

Austin, TX

May 2026 – Aug 2026

- Support autonomous mobile robot fleet commissioning and deployment in a manufacturing environment.
- Repaired and debugged AMR, PLC, and LiDAR scanner issues to support production operations.
- Developed scripts to auto-configure and push updates across the AMR fleet.

Rivian

Automation Controls Engineering Intern

Robotics simulation and virtual commissioning R2

Normal, IL

Jan 2025 – Aug 2025

- Developed full-scope production line simulations in Process Simulate.
- Supported design reviews for simulation and virtual commissioning of Rivian R2 production lines.
- Assisted the integration of a tool used to auto-generate collision zones in Process Simulate.
- Solely developed in a Unity simulation plugin realvirtual.io as a lower-cost alternative to Process Simulate.

Tesla

Automation Controls Engineering Intern

Body-in-White Robotics & Production Equipment Integration for Models 3 & Y

Fremont, CA

Jan 2024 – Aug 2024

- Oversaw and supported contract work and inventory for Body-in-White production equipment integration.
- Supported the integration of new production controls and processes to ramp up production.
- Implemented manufacturing electronics, HMIs, and PLC programs throughout multiple production lines.
- Completed training in Fanuc/KUKA (101–301), AC/DC circuits, multimeter usage, pneumatics, and PLC programming.

Dakota Integrated Systems

Automation Controls Engineering Intern

Tier 1 supplier for the Ford Expedition and Lincoln Navigator

Chicago, IL

May 2023 – Aug 2023

- Assisted sustainability practices for all process automation systems to ensure quality and continuity of design.
- Troubleshoot and maintain network performance, connectivity, and hardware to mitigate future failures.
- Developed and updated standard work instructions for all manufacturing assembly lines and stations.

Roeslein & Associates

Electrical & Instrument Designer Intern

Modular fabrication of biogas collection systems on farms domestically

Saint Louis, MO

May 2022 – Aug 2022

TECHNICAL SKILLS

Languages: C#/.NET, C++, Python

PLC and Robot Platforms: Allen-Bradley, Siemens, Kuka, Fanuc

Software: Git, Process Simulate, Isaac Lab, ROS, Unity, Ignition, Linux, MuJoCo, Docker

Key Courses: Computer Vision, Computer Animation, Reinforcement Learning, Deep Learning, Machine Learning